University Management System – README

Introduction

University Management System is a Java-based application derived from Object-Oriented Programming. It provides core university functions, including managing courses, registering students, and grading. The system facilitates three types of users:

Students: The functionality includes viewing, registering for courses, dropping courses, tracking their academic progress, and submitting complaints.

Professors: The functionality includes managing and updating course details as well as viewing students enrolled.

Administrators: It can course manage, student records, assign professors to courses, and complaints too.

The project abides by all of the major principles of OOPs like encapsulation, inheritance, and polymorphism such that the codebase formed will be modular and maintainable as well.

How to Execute the Code

Run the Main Class: Right click over the class UniversityManagementSystem (which includes a main method) and click run. Then the console-based user interface would start

Login: Use the default credentials below to login as an Administrator, Professor, or Student

Default User Credentials   For testing and demonstration purposes use the following pre-configured users: &

Admin

Username: aditya

Password: aditya

Professors

Professor 1:

Username: prof1

Password: password1

Professor 2:

Username: prof2

Password: password2

Students

Student 1:

Username: student1

Password: password1

Student 2:

Username: student2

Password: password2

Student 3:

Username: student3

Password: password3

Menu Navigation: Once you log into the application, you will be provided with a menu-based page.

Depending on the login type, whether Admin, Professor, or Student, you will be presented with the relevant options

System Functionalities

The following are functionalities that can be found in the system:

Student Functionalities:

View Available Courses: The student can view all the courses open in a semester, complete with course code, title, instructor, credits, prerequisites, and timings.

Add Courses: The students can add courses to their current semester. Prerequisites should be met along with not taking 20 credit hours in a semester.

Drop Courses: Students can drop any course in the running semester.

Weekly Schedule: Student can view the schedule for the week of courses, along with timings and meeting locations.

Academic Progress Tracking: The student can see completed courses, grades, and the calculation of their SGPA and CGPA.

Raise Complaints: Students can raise complaints against scheduling conflicts or any other issue, which will be dealt by the admin.

Professor Functionalities:

Courses Management: Professors can see and update the details of the course, such as syllabus, timings, prerequisites, credits, etc. But they can't add a new course or delete an existing one.

View Enrolled Students: Professors will be able to view a list of enrolled students in their courses in addition to the academic standing and contact information for students.

Admin Functionalities:

Manage Course Catalog: Admin can add, delete and view courses in the course catalog.

Manage Student Records: Admins can update information related to students as well as grades and other records.

Assign Professors to Courses: Based on area of expertise and availability, admin can assign professors to courses

Resolve Complaints: Admin can see and reply back to grievances from students

Assumptions

+Semester is fixed and course registration also takes place.

+Students enroll in the fall semester and are allowed to enroll only in courses that are currently available. Prerequisites-prequisites are courses completed during previous semesters-and must be completed before enrolling in higher-level courses.

Credit System: A student enrolls in 20 credits per semester. All courses are graded and credit is awarded to each course with two or four credits. To avoid unnecessary complications, the line between required courses and electives is not made.

Grade and GPA Calculation: Professors grade students at the end of every semester. SGPA and CGPA are calculated as per the no. of courses completed.

Grievance Redressal: The complaint deadline will not be automated. Students can submit complaints and admins will mark them as resolved.

OOP Concepts

1. **Encapsulation:**

Classes Student, Professor, and Course encapsulate data, through use of private fields and public methods that protect the integrity of the data.

2. **Inheritance**:

The classes Student, Professor, and Admin inherit the base class User. This enables shared attributes such as username, password to be used while adding specific functionalities for each user type

3. **Polymorphism**:

Method Overriding: The login() method is overridden in the subclasses(Student, Professor, Admin) for specific functionality about each type of user.

Dynamic Method Dispatch: The system calls the right login behavior at run time based on the type of user logging in

4. **Abstraction**:

Users interact with high-level methods, without bothering to know the implementation details. In a simple example, students may enroll in courses, but are not bothered to know how the course list is maintained, or whether it follows that the course prerequisites are met.

5. **Association**:

Classes Student and Course are associated. One student can enroll in several courses, and one course may have many students.

6. **Aggregation**:

A Course object points to Professor and Student objects but does not own them. Professors and students are in no way dependent on the course.

Demonstration Instructions

We will illustrate the use case with predefined users and courses.

1. Admin Actions:

Logging in with the default admin: admin, pass

Create two professors: prof1, prof2

Create three students: student1, student2, student3

Create five courses: Mathematics, Physics, Computer Science, English, Chemistry.

Assign professors to the courses:

Assign prof1 to Mathematics and Computer Science.

Assign prof2 to Physics, English, and Chemistry.

Review and address any student grievances filed later.

2. Professor Actions:

Login as Professor 1 (prof1).

Course Management: update course description of Mathematics and Computer Science.

Review the students enrolled in each course

Assign grades to the students in the last course of the semester

3. Student Actions:

Login as Student 1 (student1).

View available courses and enroll in Mathematics and Physics.

View the weekly schedule

Submit a complaint with regard to a scheduling conflict or other problem.

Drop a course if necessary.

View grades and grade progress once professors have entered grades

Definitions and Roles of Entities

Admin: Manages the students, professors, courses, and resolves complaints.

Professor: Manages the courses and assigns grades to students.

Student: Enrolls in courses, views schedules, tracks academic progress, and submits complaints.